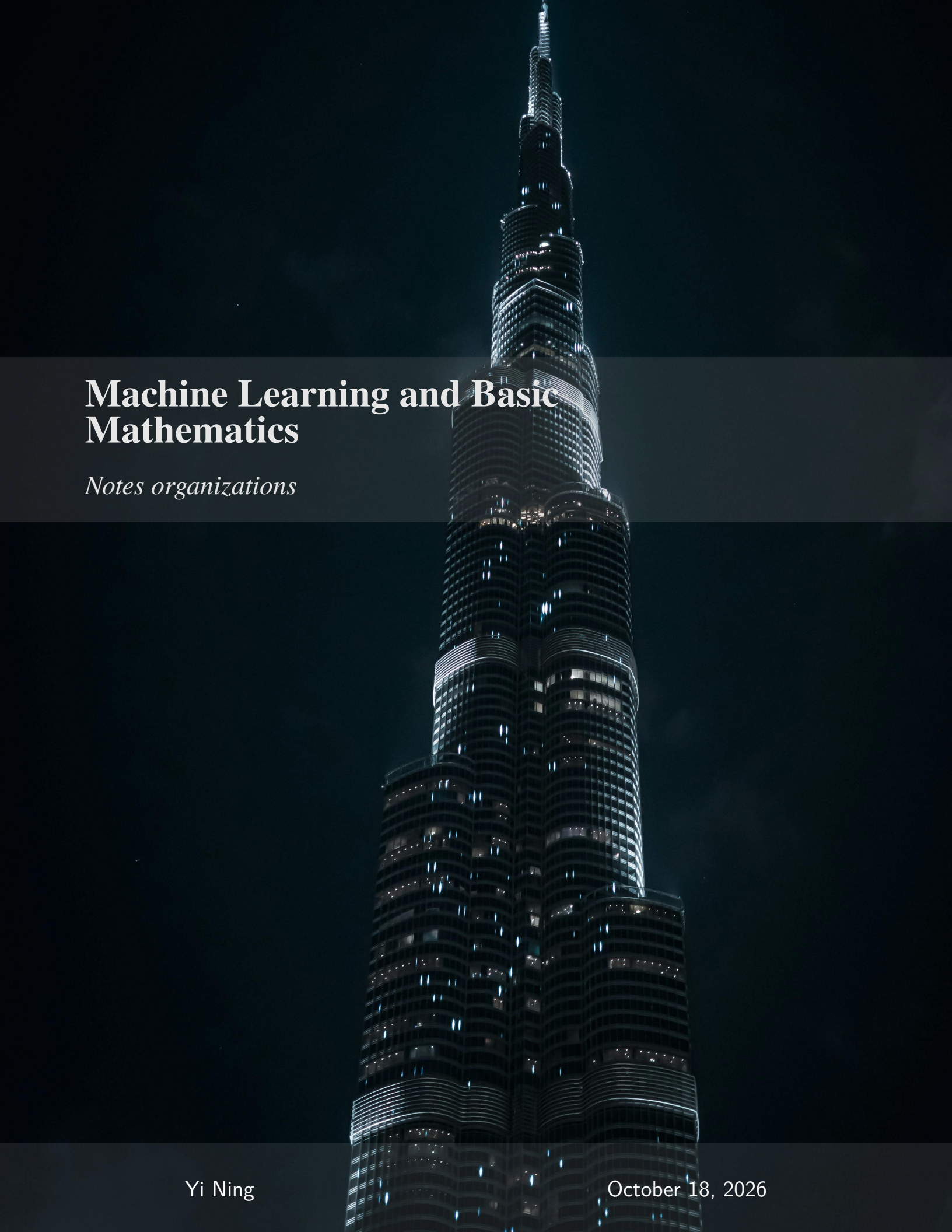


Machine Learning and Basic Mathematics

Notes organizations

Yi Ning

October 18, 2026

Contents

I Proof of Opennes Conjecture **3**

1 Notes for the Proof of Openness Conjecture **7**
1.1 Organizing the proof strategy . . . 7
1.1.1 The case 1 . . . 7
1.2 Some topics may be benifit for the future research . . . 10
1.2.1 Multiplier ideal sheaf $\mathcal{J}(\varphi)$ . . . 10
1.2.2 Plurisubharmonic functions . . . 10
1.2.3 Positive Currents . . . 11
1.3 Perverse Sheaf and Intersection Cohomology . . . 12

II Proof of Opennes Conjecture **13**

2 Notes for the Proof of Openness Conjecture **17**
2.1 Organizing the proof strategy . . . 17
2.1.1 The case 1 . . . 17
2.2 Some topics may be benifit for the future research . . . 20
2.2.1 Multiplier ideal sheaf $\mathcal{J}(\varphi)$ . . . 20
2.2.2 Plurisubharmonic functions . . . 20
2.2.3 Positive Currents . . . 21
2.3 Perverse Sheaf and Intersection Cohomology . . . 22

III Proof of Opennes Conjecture **23**

3 Notes for the Proof of Openness Conjecture **27**
3.1 Organizing the proof strategy . . . 27
3.1.1 The case 1 . . . 27
3.2 Some topics may be benifit for the future research . . . 30
3.2.1 Multiplier ideal sheaf $\mathcal{J}(\varphi)$ . . . 30
3.2.2 Plurisubharmonic functions . . . 30
3.2.3 Positive Currents . . . 31
3.3 Perverse Sheaf and Intersection Cohomology . . . 32

IV Proof of Opennes Conjecture **33**

4 Notes for the Proof of Openness Conjecture **37**
4.1 Organizing the proof strategy 37
4.1.1 The case 1 37
4.2 Some topics may be benifit for the future research 40
4.2.1 Multiplier ideal sheaf $\mathcal{J}(\varphi)$ 40
4.2.2 Plurisubharmonic functions 40
4.2.3 Positive Currents 41
4.3 Perverse Sheaf and Intersection Cohomology 42

Bibliography **43**

5 Appendix **45**
5.1 45
5.2 46
5.3 47



I
Proof of Opennes Conjecture

1

Notes for the Proof of Openness Conjecture

1.1. Organizing the proof strategy

1.1.1. The case 1

Lemma 1 (Lemma 2.3 (see [8]). *; see also [22]).] Let $F \in \mathcal{O}_n$ and $g_1, \ldots, g_s \in \mathcal{O}_n$ be germs of holomorphic functions vanishing at the origin $o \in \mathbb{C}^n$. Assume that for any given neighborhood of o, $|F| \leq C\,|(g_1, \ldots, g_s)|$ does not hold for any constant C. Then there exists a germ of an analytic curve γ through o, satisfying $\gamma \cap \{F = 0\} \subseteq \{o\}$, such that $\frac{g_i}{F}\big|_\gamma$ is holomorphic on $\gamma \backslash o$ with 2 for any $i \in \{1, \ldots, s\}$, where $\frac{\widetilde{g_i}}{F}$ is the holomorphic extension of $\frac{g_i}{F}$ from $\gamma \backslash o$ to γ.*

Theorem 2 (cf [1]). *Let φ be a negative plurisubharmonic function on the unit polydisc $\Delta^n \in \mathbb{C}^n$. Suppose F is a holomorphic function on Δ^n, which satisfies*

$$\int_{\Delta^n} |F|^2 e^{-\varphi} d\lambda_n < +\infty,$$

where $d\lambda_n$ is the Lebesgue measure on $\mathbb{C}^n$. Then for some $r \in (0, 1)$, there exists a number $p > 1$ such that

$$\int_{\Delta_r^n} |F|^2 e^{-p\varphi} d\lambda_n < +\infty.$$

Proof. As

$$\int_{\Delta' \times \Delta''} |F_j|^2 e^{-p_j \varphi} d\lambda_n \leqslant C \int_{H_j} |F|^2 e^{-p_j \varphi} d\lambda_{n-1}$$

and we already known that[1]

$$\int_{H_j} |F|^2 e^{-p_j \varphi} d\lambda_{n-1} \leqslant 2 \int_{H_j} |F|^2 e^{-\varphi} d\lambda_{n-1} = o\left(\frac{1}{|a_j|^2}\right),$$

thus we have

$$\begin{aligned}\int_{\Delta' \times \Delta''} |F_j|^2 e^{-p_j \varphi} d\lambda_n &\leqslant C \int_{H_j} |F|^2 e^{-p_j \varphi} d\lambda_{n-1} \\ &\leqslant C \int_{H_j} |F|^2 e^{-\varphi} d\lambda_{n-1} = o\left(\frac{1}{|a_j|^2}\right).\end{aligned}$$

1.1 Organizing the proof strategy 7
1.1.1 The case 1 7
1.2 Some topics may be benifit for the future research 10
1.2.1 Multiplier ideal sheaf $\mathcal{J}(\varphi)$ 10
1.2.2 Plurisubharmonic functions 10
1.2.3 Positive Currents . . . 11
1.3 Perverse Sheaf and Intersection Cohomology . . . 12

[1] see (3.7) in original paper.

Then by $F_j|_{H_j} = F|_{H_j}$ and negativeness of φ, we have

$$\int_{\Delta'\times\Delta''} |F_j|^2 d\lambda_n \leqslant \int_{\Delta'\times\Delta''} |F_j|^2 e^{-p_j\varphi} d\lambda_n = o\left(\frac{1}{|a_j|^2}\right),$$

which is to say

$$\int_{\Delta'\times\Delta''} |F_j|^2 d\lambda_n = o\left(\frac{1}{|a_j|^2}\right).$$

We choose r_0 small enough such that[2]

$$\int_{\Delta_{r_0}} |F|^2 e^{-\varphi} d\lambda_1 < +\infty.$$

Step 1: Theorem 2 for the dimension one case. We first prove Theorem 2 for the dimension one case, which is elementary but instructive. Our proof for the general dimension case is quite similar. Actually the proofs for both cases are parallel.

We choose r_0 small enough such that $\int_{\Delta_{r_0}} |F|^2 e^{-\varphi} d\lambda_1 < +\infty$[3]. Then there exist complex numbers $a_j \to 0$ $(j \to +\infty)$ such that

$$\left|F\left(a_j\right)\right|^2 e^{-\varphi(a_j)} = o\left(\frac{1}{\left|a_j\right|^2}\right).$$

As $\left|F\left(a_j\right)\right|^2 e^{-\varphi(a_j)} < +\infty$, then one can find $p_j > 1$ small enough such that[4]

$$\left|F\left(a_j\right)\right|^2 e^{-p_j\varphi(a_j)} \leq 2\left|F\left(a_j\right)\right|^2 e^{-\varphi(a_j)} = o\left(\frac{1}{\left|a_j\right|^2}\right). \tag{1.1}$$

Using movable (respect to a_j) the Ohsawa-Takegoshi L^2 extension theorem on Δ, we obtain holomorphic functions F_j on Δ such that $F_j\big|_{a_j} = F\left(a_j\right)$ and[5]

$$\int_{\Delta} \left|F_j\right|^2 e^{-p_j\varphi} d\lambda_1 \leq \mathbf{C}\left|F\left(a_j\right)\right|^2 e^{-p_j\varphi(a_j)}, \tag{1.2}$$

where $\mathbf{C}$ is a universal constant.

By inequality (4.1) and negativeness of φ, we obtain that

$$\int_{\Delta} \left|F_j\right|^2 d\lambda_1 = o\left(\frac{1}{\left|a_j\right|^2}\right). \tag{1.3}$$

By contradiction, assume that Theorem 2 does not hold for $n = 1$; that is to say, $\int_{\Delta_r} |F|^2 e^{-p_j\varphi} d\lambda = +\infty$ for any $r > 0$ and any $j \in \{1, 2, \ldots\}$.

Assertion. Since $\{F = 0\} \cap \Delta_{r_0} \subseteq \{o\}$, then it follows from inequality (4.2) that one can derive that F/F_j *is unbounded*. Otherwise, the boundedness would imply the finiteness of the integral of $|F|^2 e^{-p_j\varphi}$, according to inequality (4.2). This contradicts the assumption. Then there exists a holomorphic function h_j on Δ_{r_0} satisfying

[2] The reason is that from the assumption of theorem 2, we have known that F is a holomorphic function on Δ^n, which satisfies

$$\int_{\Delta^n} |F|^2 e^{-\varphi} d\lambda_n < +\infty,$$

thus we can find a small enough r_0 such that

$$\int_{\Delta_{r_0}} |F|^2 e^{-\varphi} d\lambda_1 < +\infty.$$

[3] It has already been explained above.

[4] As

$$\begin{aligned} & e^{-p_j\varphi(a_j)} \\ &= [e^{-\varphi(a_j)}]^{p_j} \\ &= e^{-\varphi(a_j)} \cdot [e^{-\varphi(a_j)}]^{p_j-1}, \end{aligned}$$

$p_j > 1$ $(p_j - 1 > 0)$ and the negativeness of φ $(-\varphi > 0)$, there exists p_j small enough such that $p_j - 1 \leqslant \ln 2$, thus we have

$$[e^{-\varphi(a_j)}]^{p_j-1} \leqslant e^{p_j-1} \leqslant 2.$$

[5] This is because $|F(a_j)|^2 \cdot e^{-p_j(a_j)}$ is a constant.

(1) $F_j\big|_{\Delta_{r_0}} = F|_{\Delta_{r_0}} h_j$,
(2) $h_j(o) = 0$,
(3) $h_j\left(a_j\right) = 1$.

According to Lemma 2.1[6], it follows that

$$\frac{C_1}{\left|a_j\right|^2} \leq \int_{\Delta_{r_0}} \left|F_j\right|^2 d\lambda_1,$$

which contradicts equality (4.3), where $C_1 > 0$ is independent of j. We have thus proved Theorem 2 for $n = 1$.

Step 2: Theorem 2 for n. By induction on the dimension n, one may assume that Theorem 2 holds for $n - 1$.

We prove Theorem 2 for the general dimension n by contradiction. Assume that Theorem 2 for n is not true, therefore, for some negative psh function φ, there exists a holomorphic function F such that

$$\int_{\Delta_{r_0}^n} |F|^2 e^{-\varphi} d\lambda_n < +\infty, \tag{1.4}$$

for some $r_0 > 0$, and

$$\int_{\Delta_r^n} |F|^2 e^{-p\varphi} d\lambda_n = +\infty, \tag{1.5}$$

for any $r \in (0, r_0)$ and $p > 1$. That is to say, the germ of the holomorphic function F is in $\mathcal{I}(\varphi)_o$ but not in $\mathcal{I}_+(\varphi)_o$.

By Lemma 2.4[7] and equality 4.5, it follows that there exists a germ of an analytic curve γ through o satisfying $\left\{F|_\gamma = 0\right\} \subseteq \{o\}$ such that for any germ of holomorphic function g in $\mathcal{I}_+(\varphi)_o$, there exists a holomorphic function h_g on γ satisfying

$$h_g(o) = 0 \quad \text{and} \quad g|_\gamma = F|_\gamma h_g. \tag{1.6}$$

This plays a similar role with the assertion in the proof for dimension 1.

Using the local parametrization of γ (see [5]), without loss of generality, one may assume that γ and $\Delta' \times \Delta''$ are as in Section 2.1.

By inequality (4.4), it follows that there exist hyperplanes $H_j := H_{a_j} = \left\{z' = a_j\right\}$ that satisfy $a_j \to 0 (j \to \infty)$ and

$$\int_{H_j} |F|^2 e^{-\varphi} d\lambda_{n-1} = o\left(\frac{1}{\left|a_j\right|^2}\right). \tag{1.7}$$

According to *the induction assumption* and *the Lebesgue dominated convergence theorem*, it follows that for any given j, there exists a small enough $p_j > 1$ such that[8]

[6] Let $f \not\equiv 0$ be a holomorphic function on the disc Δ_r of radius r containing the origin o in $\mathbb{C}$. Let h_a be a holomorphic function on Δ_r, which satisfies $h_a(o) = 0$ and $h_a(b) = 1$ for any $b^k = a$ (k is a positive integer), where $a \in \Delta_r$ whose norm is small enough. Then we have

$$\int_{\Delta_r} |f|^2 |h_a|^2 d\lambda_1 > C_1 |a|^{-2}$$

where C_1 is a positive constant independent of a and h_a.

[7] Assume that $F \in \mathcal{O}_n$ is a holomorphic function on some neighborhood V of o that is not a germ of

$$\mathcal{I}\left(\psi_{j_1}\right)_o = \left(\cup_{j=1}^{\infty} \mathcal{I}\left(\psi_j\right)\right)_o.$$

Then there exists a germ of an analytic curve (γ, o) such that

$$\left.\frac{\widetilde{g \circ \gamma}}{F \circ \gamma}\right|_o = 0$$

holds for any germ (g, o) of $\mathcal{I}\left(\psi_{j_1}\right)_o$, where $\frac{\widetilde{g\circ\gamma}}{F\circ\gamma}$ is the holomorphic extension of $\frac{g\circ\gamma}{F\circ\gamma}$ from $\gamma \backslash o$ to γ.

[8] It may be the reason that from the truth of one case in (4.1), we can obtain that when $a_j \to 0$, we have (4.7), thus for any given j, by dominated convergence theorem, (4.8) is true.

$$\int_{H_j}|F|^2e^{-p_j\varphi}d\lambda_{n-1}\leq 2\int_{H_j}|F|^2e^{-\varphi}d\lambda_{n-1}=o\left(\frac{1}{|a_j|^2}\right). \tag{1.8}$$

Using movably (respect to $j\longrightarrow\infty$) *the Ohsawa-Takegoshi L^2 extension theorem* on $\Delta'\times\Delta''$, we obtain a holomorphic function $F_j:=F_{a_j}$ on $\Delta'\times\Delta''$ for each j such that $F_j\big|_{H_j}=F|_{H_j}$, and

$$\int_{\Delta'\times\Delta''}\left|F_j\right|^2e^{-p_j\varphi}d\lambda_n\leq\mathbf{C}\int_{H_j}|F|^2e^{-p_j\varphi}d\lambda_{n-1},$$

where $\mathbf{C}$ is a universal constant. By inequality (4.8) and negativeness of φ, it follows that

$$\int_{\Delta'\times\Delta''}\left|F_j\right|^2d\lambda_n=o\left(\frac{1}{|a_j|^2}\right) \tag{1.9}$$

Note that $F_j\big|_{H_j}=F|_{H_j}$ and $\left(F_j,o\right)\in\mathcal{I}_+(\varphi)_o$ but $(F,o)\notin\mathcal{I}_+(\varphi)_o$. According to equality (4.6) and Lemma 2.2[9], it follows that

$$\int_{\Delta'\times\Delta''}\left|F_j\right|^2d\lambda_n\geq\frac{C_2}{|a_j|^2}$$

where $C_2>0$ is independent of j, which contradicts equality (4.9). We have thus proved Theorem 2 for n. The proof of Theorem 2 is now complete. □

[9] Let F be a holomorphic function on $\Delta'\times\Delta''$ satisfying $F|_\gamma\not\equiv 0$. Denote by hyperplane $H_a:=\{z'=a\}$ near o. Let F_a be the holomorphic extension of $F|_{H_a}$ on $\Delta'\times\Delta''$ such that there exists a holomorphic function h_a on γ satisfying

(1) $F_a|_\gamma=F|_\gamma\,h_a$,
(2) $h_a(o)=0$.

Then we have

$$\int_{\Delta'\times\Delta''}|F_a|^2\,d\lambda_n\geq\frac{C_2}{|a|^2}$$

where C_2 is a positive constant independent of a,H_a and F_a.

1.2. Some topics may be benifit for the future research

1.2.1. Multiplier ideal sheaf $\mathcal{J}(\varphi)$

Definition 3 (Multiplier ideal sheaf)**.** Multiplier ideal sheaf $\mathcal{J}(\varphi)$ is considered as the subsheaf of germs of holomorphic functions f such that $|f|^2e^{-2\varphi}$ is locally integrable (/summable).

It is worth to point out that it is a *coherent algebraic sheaf* over X and satisfies

$$H^q(X,K_X\otimes L\otimes\mathcal{J}(\varphi))=0$$

for all $q\geq 1$ if the curvature of L is positive in the sense of current.

The two efficient ways to solve the singularities are *Multiplier ideal sheaf* by Nadel (Which can be seen as a generalization of Kawamata-Viehweg's vanishing theorem.) and *The theory of positive currents* by Lelong. Currents can be seen as generalization of **algebraic cycles**.

1.2.2. Plurisubharmonic functions

Definition 4 (Plurisubharmonic functions)**.** A function $u:\Omega\longrightarrow[-\infty,+\infty[$ defined on an open subset $\Omega\subset\mathbb{C}^n$ is said to be plurisubharmonic (psh for short) if

(a) u is upper semicontinuous;

[10] A vital characteristic, which is equivalent to the integrability of the inegral.

(b) for every complex line $L \subset \mathbb{C}^n$, $u_{\restriction \Omega \cap L}$ is subharmonic on $\Omega \cap L$, that is, for all $a \in \Omega$ and $\xi \in \mathbb{C}^n$ with $|\xi| < d(a, \complement\Omega)$, the function u satisfies the mean value inequality[10]

$$u(a) \leqslant \frac{1}{2\pi} \int_0^{2\pi} u\left(a + e^{i\theta}\xi\right) d\theta.$$

The set of psh functions on Ω is denoted by $\mathrm{Psh}(\Omega)$.

Theorem 5. *A function $u \in C^2(\Omega, \mathbb{R})$ is psh if and only if the hermitian form*

$$Hu(a)(\xi) = \sum_{1 \leqslant j,k \leqslant n} \frac{{}^2u}{z_j \bar{z}_k(a)} \xi_j \bar{\xi}_k$$

is semi-positive at every point $a \in \Omega$.

1.2.3. Positive Currents

A current of degree q on an oriented differentiable manifold M is simply a differential q-form Θ with distribution coefficients. The space of currents of degree q over M will be denoted by $\mathscr{D}'^q(M)$. Alternatively, a current of degree q can be seen as an element Θ in the dual space $\mathscr{D}'_p(M) := (\mathscr{D}^p(M))'$ of the space $\mathscr{D}^p(M)$ of smooth differential forms of degree $p = \dim M - q$ with compact support; the duality pairing is given by

$$\langle \Theta, \alpha \rangle = \int_M \Theta \wedge \alpha, \quad \alpha \in \mathscr{D}^p(M). \tag{1.10}$$

A basic example is the current of integration $[S]$ over a compact oriented submanifold S of M :

$$\langle [S], \alpha \rangle = \int_S \alpha, \quad \deg \alpha = p = \dim_{\mathbb{R}} S \tag{1.11}$$

Then $[S]$ is a current with measure coefficients, and Stokes' formula shows that $d[S] = (-1)^{q-1}[S]$, in particular $d[S] = 0$ if S has no boundary. Because of this example, the integer p is said to be the dimension of Θ when $\Theta \in \mathscr{D}'_p(M)$. The current Θ is said to be closed if $d\Theta = 0$.

On a complex manifold X, we have similar notions of bidegree and bidimension; as in the real case, we denote by

$$\mathscr{D}'^{p,q}(X) = \mathscr{D}'_{n-p,n-q}(X), \quad n = \dim X, \tag{1.12}$$

the space of currents of bidegree (p, q) and bidimension $(n-p, n-q)$ on X. According to [2], a current Θ of bidimension (p, p) is said to be (weakly) positive if for every choice of smooth (1,0)-forms $\alpha_1, \ldots, \alpha_p$ on X the distribution

$$\Theta \wedge (i\alpha_1 \wedge \bar{\alpha}_1) \wedge \ldots \wedge (i\alpha_p \wedge \bar{\alpha}_p)$$

is a positive measure.

[2] P. Lelong. "Intégration sur un ensemble analytique complexe". In: *Bulletin de la Société Mathématique de France* 85 (1957), pp. 239–262. DOI: 10.24033/bsmf.1488. URL: http://www.numdam.org/articles/10.24033/bsmf.1488/

1.3. Perverse Sheaf and Intersection Cohomology

- test
  - testing
    - tested
  - testii
- test2
- test3

1. test
   (a). testing
      I. tested
   (b). testii
2. test2
3. test3



II
Proof of Opennes Conjecture

2

Notes for the Proof of Openness Conjecture

2.1. Organizing the proof strategy

2.1.1. The case 1

Lemma 1 (Lemma 2.3 (see [8]). *; see also [22]).] Let $F \in \mathcal{O}_n$ and $g_1, \ldots, g_s \in \mathcal{O}_n$ be germs of holomorphic functions vanishing at the origin $o \in \mathbb{C}^n$. Assume that for any given neighborhood of o, $|F| \leq C\,|(g_1, \ldots, g_s)|$ does not hold for any constant C. Then there exists a germ of an analytic curve γ through o, satisfying $\gamma \cap \{F = 0\} \subseteq \{o\}$, such that $\frac{g_i}{F}\big|_\gamma$ is holomorphic on $\gamma \backslash o$ with 2 for any $i \in \{1, \ldots, s\}$, where $\frac{\widetilde{g_i}}{F}$ is the holomorphic extension of $\frac{g_i}{F}$ from $\gamma \backslash o$ to γ.*

Theorem 2 (cf [1]). *Let φ be a negative plurisubharmonic function on the unit polydisc $\Delta^n \in \mathbb{C}^n$. Suppose F is a holomorphic function on Δ^n, which satisfies*

$$\int_{\Delta^n} |F|^2 e^{-\varphi} d\lambda_n < +\infty,$$

where $d\lambda_n$ is the Lebesgue measure on $\mathbb{C}^n$. Then for some $r \in (0, 1)$, there exists a number $p > 1$ such that

$$\int_{\Delta_r^n} |F|^2 e^{-p\varphi} d\lambda_n < +\infty.$$

Proof. As

$$\int_{\Delta' \times \Delta''} |F_j|^2 e^{-p_j \varphi} d\lambda_n \leqslant C \int_{H_j} |F|^2 e^{-p_j \varphi} d\lambda_{n-1}$$

and we already known that[11]

$$\int_{H_j} |F|^2 e^{-p_j \varphi} d\lambda_{n-1} \leqslant 2 \int_{H_j} |F|^2 e^{-\varphi} d\lambda_{n-1} = o\left(\frac{1}{|a_j|^2}\right),$$

thus we have

$$\begin{aligned}\int_{\Delta' \times \Delta''} |F_j|^2 e^{-p_j \varphi} d\lambda_n &\leqslant C \int_{H_j} |F|^2 e^{-p_j \varphi} d\lambda_{n-1} \\ &\leqslant C \int_{H_j} |F|^2 e^{-\varphi} d\lambda_{n-1} = o\left(\frac{1}{|a_j|^2}\right).\end{aligned}$$

[11] see (3.7) in original paper.

2.1 Organizing the proof strategy 17
2.1.1 The case 1 17
2.2 Some topics may be benifit for the future research 20
2.2.1 Multiplier ideal sheaf $\mathcal{J}(\varphi)$ 20
2.2.2 Plurisubharmonic functions 20
2.2.3 Positive Currents . . . 21
2.3 Perverse Sheaf and Intersection Cohomology . . . 22

Then by $F_j|_{H_j} = F|_{H_j}$ and negativeness of φ, we have

$$\int_{\Delta'\times\Delta''} |F_j|^2 d\lambda_n \leqslant \int_{\Delta'\times\Delta''} |F_j|^2 e^{-p_j\varphi} d\lambda_n = o\left(\frac{1}{|a_j|^2}\right),$$

which is to say

$$\int_{\Delta'\times\Delta''} |F_j|^2 d\lambda_n = o\left(\frac{1}{|a_j|^2}\right).$$

We choose r_0 small enough such that[12]

$$\int_{\Delta_{r_0}} |F|^2 e^{-\varphi} d\lambda_1 < +\infty.$$

Step 1: Theorem 2 for the dimension one case. We first prove Theorem 2 for the dimension one case, which is elementary but instructive. Our proof for the general dimension case is quite similar. Actually the proofs for both cases are parallel.

We choose r_0 small enough such that $\int_{\Delta_{r_0}} |F|^2 e^{-\varphi} d\lambda_1 < +\infty$[13]. Then there exist complex numbers $a_j \to 0$ $(j \to +\infty)$ such that

$$\left|F\left(a_j\right)\right|^2 e^{-\varphi(a_j)} = o\left(\frac{1}{\left|a_j\right|^2}\right).$$

As $\left|F\left(a_j\right)\right|^2 e^{-\varphi(a_j)} < +\infty$, then one can find $p_j > 1$ small enough such that[14]

$$\left|F\left(a_j\right)\right|^2 e^{-p_j\varphi(a_j)} \leq 2\left|F\left(a_j\right)\right|^2 e^{-\varphi(a_j)} = o\left(\frac{1}{\left|a_j\right|^2}\right). \tag{2.1}$$

Using movable (respect to a_j) the Ohsawa-Takegoshi L^2 extension theorem on Δ, we obtain holomorphic functions F_j on Δ such that $F_j\big|_{a_j} = F\left(a_j\right)$ and[15]

$$\int_{\Delta} \left|F_j\right|^2 e^{-p_j\varphi} d\lambda_1 \leq \mathbf{C}\left|F\left(a_j\right)\right|^2 e^{-p_j\varphi(a_j)}, \tag{2.2}$$

where $\mathbf{C}$ is a universal constant.

By inequality (4.1) and negativeness of φ, we obtain that

$$\int_{\Delta} \left|F_j\right|^2 d\lambda_1 = o\left(\frac{1}{\left|a_j\right|^2}\right). \tag{2.3}$$

By contradiction, assume that Theorem 2 does not hold for $n = 1$; that is to say, $\int_{\Delta_r} |F|^2 e^{-p_j\varphi} d\lambda = +\infty$ for any $r > 0$ and any $j \in \{1, 2, \ldots\}$.

Assertion. Since $\{F = 0\} \cap \Delta_{r_0} \subseteq \{o\}$, then it follows from inequality (4.2) that one can derive that F/F_j *is unbounded*. Otherwise, the boundedness would imply the finiteness of the integral of $|F|^2 e^{-p_j\varphi}$, according to inequality (4.2). This contradicts the assumption. Then there exists a holomorphic function h_j on Δ_{r_0} satisfying

[12] The reason is that from the assumption of theorem 2, we have known that F is a holomorphic function on Δ^n, which satisfies

$$\int_{\Delta^n} |F|^2 e^{-\varphi} d\lambda_n < +\infty,$$

thus we can find a small enough r_0 such that

$$\int_{\Delta_{r_0}} |F|^2 e^{-\varphi} d\lambda_1 < +\infty.$$

[13] It has already been explained above.

[14] As

$$\begin{aligned} & e^{-p_j\varphi(a_j)} \\ &= [e^{-\varphi(a_j)}]^{p_j} \\ &= e^{-\varphi(a_j)} \cdot [e^{-\varphi(a_j)}]^{p_j-1}, \end{aligned}$$

$p_j > 1$ $(p_j - 1 > 0)$ and the negativeness of φ $(-\varphi > 0)$, there exists p_j small enough such that $p_j - 1 \leqslant \ln 2$, thus we have

$$[e^{-\varphi(a_j)}]^{p_j-1} \leqslant e^{p_j-1} \leqslant 2.$$

[15] This is because $|F(a_j)|^2 \cdot e^{-p_j(a_j)}$ is a constant.

(1) $F_j\big|_{\Delta_{r_0}} = F|_{\Delta_{r_0}} h_j$,
(2) $h_j(o) = 0$,
(3) $h_j\left(a_j\right) = 1$.

According to Lemma 2.1[16], it follows that

$$\frac{C_1}{\left|a_j\right|^2} \leq \int_{\Delta_{r_0}} \left|F_j\right|^2 d\lambda_1,$$

which contradicts equality (4.3), where $C_1 > 0$ is independent of j. We have thus proved Theorem 2 for $n = 1$.

Step 2: Theorem 2 for n. By induction on the dimension n, one may assume that Theorem 2 holds for $n - 1$.

We prove Theorem 2 for the general dimension n by contradiction. Assume that Theorem 2 for n is not true, therefore, for some negative psh function φ, there exists a holomorphic function F such that

$$\int_{\Delta_{r_0}^n} |F|^2 e^{-\varphi} d\lambda_n < +\infty, \tag{2.4}$$

for some $r_0 > 0$, and

$$\int_{\Delta_r^n} |F|^2 e^{-p\varphi} d\lambda_n = +\infty, \tag{2.5}$$

for any $r \in (0, r_0)$ and $p > 1$. That is to say, the germ of the holomorphic function F is in $\mathcal{I}(\varphi)_o$ but not in $\mathcal{I}_+(\varphi)_o$.

By Lemma 2.4[17] and equality 4.5, it follows that there exists a germ of an analytic curve γ through o satisfying $\left\{F|_\gamma = 0\right\} \subseteq \{o\}$ such that for any germ of holomorphic function g in $\mathcal{I}_+(\varphi)_o$, there exists a holomorphic function h_g on γ satisfying

$$h_g(o) = 0 \quad \text{and} \quad g|_\gamma = F|_\gamma h_g. \tag{2.6}$$

This plays a similar role with the assertion in the proof for dimension 1.

Using the local parametrization of γ (see [5]), without loss of generality, one may assume that γ and $\Delta' \times \Delta''$ are as in Section 2.1.

By inequality (4.4), it follows that there exist hyperplanes $H_j := H_{a_j} = \left\{z' = a_j\right\}$ that satisfy $a_j \to 0 (j \to \infty)$ and

$$\int_{H_j} |F|^2 e^{-\varphi} d\lambda_{n-1} = o\left(\frac{1}{\left|a_j\right|^2}\right). \tag{2.7}$$

According to *the induction assumption* and *the Lebesgue dominated convergence theorem*, it follows that for any given j, there exists a small enough $p_j > 1$ such that[18]

[16] Let $f \not\equiv 0$ be a holomorphic function on the disc Δ_r of radius r containing the origin o in $\mathbb{C}$. Let h_a be a holomorphic function on Δ_r, which satisfies $h_a(o) = 0$ and $h_a(b) = 1$ for any $b^k = a$ (k is a positive integer), where $a \in \Delta_r$ whose norm is small enough. Then we have

$$\int_{\Delta_r} |f|^2 |h_a|^2 d\lambda_1 > C_1 |a|^{-2}$$

where C_1 is a positive constant independent of a and h_a.

[17] Assume that $F \in \mathcal{O}_n$ is a holomorphic function on some neighborhood V of o that is not a germ of

$$\mathcal{I}\left(\psi_{j_1}\right)_o = \left(\cup_{j=1}^{\infty} \mathcal{I}\left(\psi_j\right)\right)_o.$$

Then there exists a germ of an analytic curve (γ, o) such that

$$\left.\frac{\widetilde{g \circ \gamma}}{F \circ \gamma}\right|_o = 0$$

holds for any germ (g, o) of $\mathcal{I}\left(\psi_{j_1}\right)_o$, where $\frac{\widetilde{g\circ\gamma}}{F\circ\gamma}$ is the holomorphic extension of $\frac{g\circ\gamma}{F\circ\gamma}$ from $\gamma \backslash o$ to γ.

[18] It may be the reason that from the truth of one case in (4.1), we can obtain that when $a_j \to 0$, we have (4.7), thus for any given j, by dominated convergence theorem, (4.8) is true.

$$\int_{H_j} |F|^2 e^{-p_j\varphi} d\lambda_{n-1} \leq 2\int_{H_j} |F|^2 e^{-\varphi} d\lambda_{n-1} = o\left(\frac{1}{|a_j|^2}\right). \tag{2.8}$$

Using movably (respect to $j \longrightarrow \infty$) *the Ohsawa-Takegoshi L^2 extension theorem* on $\Delta' \times \Delta''$, we obtain a holomorphic function $F_j := F_{a_j}$ on $\Delta' \times \Delta''$ for each j such that $F_j\big|_{H_j} = F|_{H_j}$, and

$$\int_{\Delta'\times\Delta''} \left|F_j\right|^2 e^{-p_j\varphi} d\lambda_n \leq \mathbf{C}\int_{H_j} |F|^2 e^{-p_j\varphi} d\lambda_{n-1},$$

where $\mathbf{C}$ is a universal constant. By inequality (4.8) and negativeness of φ, it follows that

$$\int_{\Delta'\times\Delta''} \left|F_j\right|^2 d\lambda_n = o\left(\frac{1}{|a_j|^2}\right) \tag{2.9}$$

Note that $F_j\big|_{H_j} = F|_{H_j}$ and $(F_j, o) \in \mathcal{I}_+(\varphi)_o$ but $(F, o) \notin \mathcal{I}_+(\varphi)_o$. According to equality (4.6) and Lemma 2.2[19], it follows that

$$\int_{\Delta'\times\Delta''} \left|F_j\right|^2 d\lambda_n \geq \frac{C_2}{|a_j|^2}$$

where $C_2 > 0$ is independent of j, which contradicts equality (4.9). We have thus proved Theorem 2 for n. The proof of Theorem 2 is now complete. □

[19] Let F be a holomorphic function on $\Delta' \times \Delta''$ satisfying $F|_\gamma \not\equiv 0$. Denote by hyperplane $H_a := \{z' = a\}$ near o. Let F_a be the holomorphic extension of $F|_{H_a}$ on $\Delta' \times \Delta''$ such that there exists a holomorphic function h_a on γ satisfying

(1) $F_a|_\gamma = F|_\gamma\, h_a$,
(2) $h_a(o) = 0$.

Then we have

$$\int_{\Delta'\times\Delta''} |F_a|^2 d\lambda_n \geq \frac{C_2}{|a|^2}$$

where C_2 is a positive constant independent of a, H_a and F_a.

2.2. Some topics may be benifit for the future research

2.2.1. Multiplier ideal sheaf $\mathcal{J}(\varphi)$

Definition 3 (Multiplier ideal sheaf)**.** Multiplier ideal sheaf $\mathcal{J}(\varphi)$ is considered as the subsheaf of germs of holomorphic functions f such that $|f|^2 e^{-2\varphi}$ is locally integrable (/summable).

It is worth to point out that it is a *coherent algebraic sheaf* over X and satisfies

$$H^q(X, K_X \otimes L \otimes \mathcal{J}(\varphi)) = 0$$

for all $q \geq 1$ if the curvature of L is positive in the sense of current.

The two efficient ways to solve the singularities are *Multiplier ideal sheaf* by Nadel (Which can be seen as a generalization of Kawamata-Viehweg's vanishing theorem.) and *The theory of positive currents* by Lelong. Currents can be seen as generalization of **algebraic cycles**.

2.2.2. Plurisubharmonic functions

Definition 4 (Plurisubharmonic functions)**.** A function $u : \Omega \longrightarrow [-\infty, +\infty[$ defined on an open subset $\Omega \subset \mathbb{C}^n$ is said to be plurisubharmonic (psh for short) if

(a) u is upper semicontinuous;

[20] A vital characteristic, which is equivalent to the integrability of the inegral.

(b) for every complex line $L \subset \mathbb{C}^n$, $u_{\restriction \Omega \cap L}$ is subharmonic on $\Omega \cap L$, that is, for all $a \in \Omega$ and $\xi \in \mathbb{C}^n$ with $|\xi| < d(a, \complement\Omega)$, the function u satisfies the mean value inequality[20]

$$u(a) \leqslant \frac{1}{2\pi} \int_0^{2\pi} u\left(a + e^{i\theta}\xi\right) d\theta.$$

The set of psh functions on Ω is denoted by $\mathrm{Psh}(\Omega)$.

Theorem 5. *A function $u \in C^2(\Omega, \mathbb{R})$ is psh if and only if the hermitian form*

$$Hu(a)(\xi) = \sum_{1 \leqslant j,k \leqslant n} \frac{{}^2u}{z_j \bar{z}_k(a)} \xi_j \bar{\xi}_k$$

is semi-positive at every point $a \in \Omega$.

2.2.3. Positive Currents

A current of degree q on an oriented differentiable manifold M is simply a differential q-form Θ with distribution coefficients. The space of currents of degree q over M will be denoted by $\mathscr{D}'^q(M)$. Alternatively, a current of degree q can be seen as an element Θ in the dual space $\mathscr{D}'_p(M) := (\mathscr{D}^p(M))'$ of the space $\mathscr{D}^p(M)$ of smooth differential forms of degree $p = \dim M - q$ with compact support; the duality pairing is given by

$$\langle \Theta, \alpha \rangle = \int_M \Theta \wedge \alpha, \quad \alpha \in \mathscr{D}^p(M). \tag{2.10}$$

A basic example is the current of integration $[S]$ over a compact oriented submanifold S of M :

$$\langle [S], \alpha \rangle = \int_S \alpha, \quad \deg \alpha = p = \dim_{\mathbb{R}} S \tag{2.11}$$

Then $[S]$ is a current with measure coefficients, and Stokes' formula shows that $d[S] = (-1)^{q-1}[S]$, in particular $d[S] = 0$ if S has no boundary. Because of this example, the integer p is said to be the dimension of Θ when $\Theta \in \mathscr{D}'_p(M)$. The current Θ is said to be closed if $d\Theta = 0$.

On a complex manifold X, we have similar notions of bidegree and bidimension; as in the real case, we denote by

$$\mathscr{D}'^{p,q}(X) = \mathscr{D}'_{n-p,n-q}(X), \quad n = \dim X, \tag{2.12}$$

the space of currents of bidegree (p, q) and bidimension $(n-p, n-q)$ on X. According to [2], a current Θ of bidimension $(\ p, p\ )$ is said to be (weakly) positive if for every choice of smooth $(\ 1{,}0\ )$-forms $\alpha_1, \ldots, \alpha_p$ on X the distribution

$$\Theta \wedge (i\alpha_1 \wedge \bar{\alpha}_1) \wedge \ldots \wedge (i\alpha_p \wedge \bar{\alpha}_p)$$

is a positive measure.

[2] P. Lelong. "Intégration sur un ensemble analytique complexe". In: *Bulletin de la Société Mathématique de France* 85 (1957), pp. 239–262. DOI: 10.24033/bsmf.1488. URL: http://www.numdam.org/articles/10.24033/bsmf.1488/

2.3. Perverse Sheaf and Intersection Cohomology

- test
  - testing
    - tested
  - testii
- test2
- test3

1. test
   (a). testing
      I. tested
   (b). testii
2. test2
3. test3



III
Proof of Opennes Conjecture

3

Notes for the Proof of Openness Conjecture

3.1. Organizing the proof strategy

3.1.1. The case 1

Lemma 1 (Lemma 2.3 (see [8]). *; see also [22]).] Let $F \in \mathcal{O}_n$ and $g_1, \ldots, g_s \in \mathcal{O}_n$ be germs of holomorphic functions vanishing at the origin $o \in \mathbb{C}^n$. Assume that for any given neighborhood of o, $|F| \leq C\,|(g_1, \ldots, g_s)|$ does not hold for any constant C. Then there exists a germ of an analytic curve γ through o, satisfying $\gamma \cap \{F = 0\} \subseteq \{o\}$, such that $\frac{g_i}{F}\big|_\gamma$ is holomorphic on $\gamma \backslash o$ with 2 for any $i \in \{1, \ldots, s\}$, where $\frac{\widetilde{g_i}}{F}$ is the holomorphic extension of $\frac{g_i}{F}$ from $\gamma \backslash o$ to γ.*

Theorem 2 (cf [1]). *Let φ be a negative plurisubharmonic function on the unit polydisc $\Delta^n \in \mathbb{C}^n$. Suppose F is a holomorphic function on Δ^n, which satisfies*

$$\int_{\Delta^n} |F|^2 e^{-\varphi} d\lambda_n < +\infty,$$

where $d\lambda_n$ is the Lebesgue measure on $\mathbb{C}^n$. Then for some $r \in (0, 1)$, there exists a number $p > 1$ such that

$$\int_{\Delta_r^n} |F|^2 e^{-p\varphi} d\lambda_n < +\infty.$$

Proof. As

$$\int_{\Delta' \times \Delta''} |F_j|^2 e^{-p_j \varphi} d\lambda_n \leqslant C \int_{H_j} |F|^2 e^{-p_j \varphi} d\lambda_{n-1}$$

and we already known that[21]

$$\int_{H_j} |F|^2 e^{-p_j \varphi} d\lambda_{n-1} \leqslant 2 \int_{H_j} |F|^2 e^{-\varphi} d\lambda_{n-1} = o\left(\frac{1}{|a_j|^2}\right),$$

thus we have

$$\begin{aligned}\int_{\Delta' \times \Delta''} |F_j|^2 e^{-p_j \varphi} d\lambda_n &\leqslant C \int_{H_j} |F|^2 e^{-p_j \varphi} d\lambda_{n-1} \\ &\leqslant C \int_{H_j} |F|^2 e^{-\varphi} d\lambda_{n-1} = o\left(\frac{1}{|a_j|^2}\right).\end{aligned}$$

3.1 Organizing the proof strategy 27
3.1.1 The case 1 27
3.2 Some topics may be benifit for the future research 30
3.2.1 Multiplier ideal sheaf $\mathcal{J}(\varphi)$ 30
3.2.2 Plurisubharmonic functions 30
3.2.3 Positive Currents . . . 31
3.3 Perverse Sheaf and Intersection Cohomology . . . 32

[21] see (3.7) in original paper.

Then by $F_j|_{H_j} = F|_{H_j}$ and negativeness of φ, we have

$$\int_{\Delta'\times\Delta''} |F_j|^2 d\lambda_n \leqslant \int_{\Delta'\times\Delta''} |F_j|^2 e^{-p_j\varphi} d\lambda_n = o\left(\frac{1}{|a_j|^2}\right),$$

which is to say

$$\int_{\Delta'\times\Delta''} |F_j|^2 d\lambda_n = o\left(\frac{1}{|a_j|^2}\right).$$

We choose r_0 small enough such that[22]

$$\int_{\Delta_{r_0}} |F|^2 e^{-\varphi} d\lambda_1 < +\infty.$$

Step 1: Theorem 2 for the dimension one case. We first prove Theorem 2 for the dimension one case, which is elementary but instructive. Our proof for the general dimension case is quite similar. Actually the proofs for both cases are parallel.

We choose r_0 small enough such that $\int_{\Delta_{r_0}} |F|^2 e^{-\varphi} d\lambda_1 < +\infty$[23]. Then there exist complex numbers $a_j \to 0$ $(j \to +\infty)$ such that

$$\left|F\left(a_j\right)\right|^2 e^{-\varphi(a_j)} = o\left(\frac{1}{\left|a_j\right|^2}\right).$$

As $\left|F\left(a_j\right)\right|^2 e^{-\varphi(a_j)} < +\infty$, then one can find $p_j > 1$ small enough such that[24]

$$\left|F\left(a_j\right)\right|^2 e^{-p_j\varphi(a_j)} \leq 2\left|F\left(a_j\right)\right|^2 e^{-\varphi(a_j)} = o\left(\frac{1}{\left|a_j\right|^2}\right). \tag{3.1}$$

Using movable (respect to a_j) the Ohsawa-Takegoshi L^2 extension theorem on Δ, we obtain holomorphic functions F_j on Δ such that $F_j\big|_{a_j} = F\left(a_j\right)$ and[25]

$$\int_{\Delta} \left|F_j\right|^2 e^{-p_j\varphi} d\lambda_1 \leq \mathbf{C}\left|F\left(a_j\right)\right|^2 e^{-p_j\varphi(a_j)}, \tag{3.2}$$

where $\mathbf{C}$ is a universal constant.

By inequality (4.1) and negativeness of φ, we obtain that

$$\int_{\Delta} \left|F_j\right|^2 d\lambda_1 = o\left(\frac{1}{\left|a_j\right|^2}\right). \tag{3.3}$$

By contradiction, assume that Theorem 2 does not hold for $n = 1$; that is to say, $\int_{\Delta_r} |F|^2 e^{-p_j\varphi} d\lambda = +\infty$ for any $r > 0$ and any $j \in \{1, 2, \ldots\}$.

Assertion. Since $\{F = 0\} \cap \Delta_{r_0} \subseteq \{o\}$, then it follows from inequality (4.2) that one can derive that F/F_j *is unbounded*. Otherwise, the boundedness would imply the finiteness of the integral of $|F|^2 e^{-p_j\varphi}$, according to inequality (4.2). This contradicts the assumption. Then there exists a holomorphic function h_j on Δ_{r_0} satisfying

[22] The reason is that from the assumption of theorem 2, we have known that F is a holomorphic function on Δ^n, which satisfies

$$\int_{\Delta^n} |F|^2 e^{-\varphi} d\lambda_n < +\infty,$$

thus we can find a small enough r_0 such that

$$\int_{\Delta_{r_0}} |F|^2 e^{-\varphi} d\lambda_1 < +\infty.$$

[23] It has already been explained above.

[24] As

$$\begin{aligned} & e^{-p_j\varphi(a_j)} \\ &= [e^{-\varphi(a_j)}]^{p_j} \\ &= e^{-\varphi(a_j)} \cdot [e^{-\varphi(a_j)}]^{p_j-1}, \end{aligned}$$

$p_j > 1$ $(p_j - 1 > 0)$ and the negativeness of φ $(-\varphi > 0)$, there exists p_j small enough such that $p_j - 1 \leqslant \ln 2$, thus we have

$$[e^{-\varphi(a_j)}]^{p_j-1} \leqslant e^{p_j-1} \leqslant 2.$$

[25] This is because $|F(a_j)|^2 \cdot e^{-p_j(a_j)}$ is a constant.

(1) $F_j\big|_{\Delta_{r_0}} = F|_{\Delta_{r_0}} h_j$,
(2) $h_j(o) = 0$,
(3) $h_j\left(a_j\right) = 1$.

According to Lemma 2.1[26], it follows that

$$\frac{C_1}{\left|a_j\right|^2} \leq \int_{\Delta_{r_0}} \left|F_j\right|^2 d\lambda_1,$$

which contradicts equality (4.3), where $C_1 > 0$ is independent of j. We have thus proved Theorem 2 for $n = 1$.

Step 2: Theorem 2 for n. By induction on the dimension n, one may assume that Theorem 2 holds for $n - 1$.

We prove Theorem 2 for the general dimension n by contradiction. Assume that Theorem 2 for n is not true, therefore, for some negative psh function φ, there exists a holomorphic function F such that

$$\int_{\Delta_{r_0}^n} |F|^2 e^{-\varphi} d\lambda_n < +\infty, \tag{3.4}$$

for some $r_0 > 0$, and

$$\int_{\Delta_r^n} |F|^2 e^{-p\varphi} d\lambda_n = +\infty, \tag{3.5}$$

for any $r \in (0, r_0)$ and $p > 1$. That is to say, the germ of the holomorphic function F is in $\mathcal{I}(\varphi)_o$ but not in $\mathcal{I}_+(\varphi)_o$.

By Lemma 2.4[27] and equality 4.5, it follows that there exists a germ of an analytic curve γ through o satisfying $\left\{F|_\gamma = 0\right\} \subseteq \{o\}$ such that for any germ of holomorphic function g in $\mathcal{I}_+(\varphi)_o$, there exists a holomorphic function h_g on γ satisfying

$$h_g(o) = 0 \quad \text{and} \quad g|_\gamma = F|_\gamma h_g. \tag{3.6}$$

This plays a similar role with the assertion in the proof for dimension 1.

Using the local parametrization of γ (see [5]), without loss of generality, one may assume that γ and $\Delta' \times \Delta''$ are as in Section 2.1.

By inequality (4.4), it follows that there exist hyperplanes $H_j := H_{a_j} = \left\{z' = a_j\right\}$ that satisfy $a_j \to 0 (j \to \infty)$ and

$$\int_{H_j} |F|^2 e^{-\varphi} d\lambda_{n-1} = o\left(\frac{1}{\left|a_j\right|^2}\right). \tag{3.7}$$

According to *the induction assumption* and *the Lebesgue dominated convergence theorem*, it follows that for any given j, there exists a small enough $p_j > 1$ such that[28]

[26] Let $f \not\equiv 0$ be a holomorphic function on the disc Δ_r of radius r containing the origin o in $\mathbb{C}$. Let h_a be a holomorphic function on Δ_r, which satisfies $h_a(o) = 0$ and $h_a(b) = 1$ for any $b^k = a$ (k is a positive integer), where $a \in \Delta_r$ whose norm is small enough. Then we have

$$\int_{\Delta_r} |f|^2 |h_a|^2 d\lambda_1 > C_1 |a|^{-2}$$

where C_1 is a positive constant independent of a and h_a.

[27] Assume that $F \in \mathcal{O}_n$ is a holomorphic function on some neighborhood V of o that is not a germ of

$$\mathcal{I}\left(\psi_{j_1}\right)_o = \left(\cup_{j=1}^{\infty} \mathcal{I}\left(\psi_j\right)\right)_o.$$

Then there exists a germ of an analytic curve (γ, o) such that

$$\left.\frac{\widetilde{g \circ \gamma}}{F \circ \gamma}\right|_o = 0$$

holds for any germ (g, o) of $\mathcal{I}\left(\psi_{j_1}\right)_o$, where $\frac{\widetilde{g\circ\gamma}}{F\circ\gamma}$ is the holomorphic extension of $\frac{g\circ\gamma}{F\circ\gamma}$ from $\gamma \backslash o$ to γ.

[28] It may be the reason that from the truth of one case in (4.1), we can obtain that when $a_j \to 0$, we have (4.7), thus for any given j, by dominated convergence theorem, (4.8) is true.

$$\int_{H_j} |F|^2 e^{-p_j\varphi} d\lambda_{n-1} \leq 2\int_{H_j} |F|^2 e^{-\varphi} d\lambda_{n-1} = o\left(\frac{1}{|a_j|^2}\right). \tag{3.8}$$

Using movably (respect to $j \longrightarrow \infty$) *the Ohsawa-Takegoshi L^2 extension theorem* on $\Delta' \times \Delta''$, we obtain a holomorphic function $F_j := F_{a_j}$ on $\Delta' \times \Delta''$ for each j such that $F_j\big|_{H_j} = F|_{H_j}$, and

$$\int_{\Delta'\times\Delta''} \left|F_j\right|^2 e^{-p_j\varphi} d\lambda_n \leq \mathbf{C}\int_{H_j} |F|^2 e^{-p_j\varphi} d\lambda_{n-1},$$

where $\mathbf{C}$ is a universal constant. By inequality (4.8) and negativeness of φ, it follows that

$$\int_{\Delta'\times\Delta''} \left|F_j\right|^2 d\lambda_n = o\left(\frac{1}{|a_j|^2}\right) \tag{3.9}$$

Note that $F_j\big|_{H_j} = F|_{H_j}$ and $(F_j, o) \in \mathcal{I}_+(\varphi)_o$ but $(F, o) \notin \mathcal{I}_+(\varphi)_o$. According to equality (4.6) and Lemma 2.2[29], it follows that

$$\int_{\Delta'\times\Delta''} \left|F_j\right|^2 d\lambda_n \geq \frac{C_2}{|a_j|^2}$$

where $C_2 > 0$ is independent of j, which contradicts equality (4.9). We have thus proved Theorem 2 for n. The proof of Theorem 2 is now complete. □

[29] Let F be a holomorphic function on $\Delta' \times \Delta''$ satisfying $F|_\gamma \not\equiv 0$. Denote by hyperplane $H_a := \{z' = a\}$ near o. Let F_a be the holomorphic extension of $F|_{H_a}$ on $\Delta' \times \Delta''$ such that there exists a holomorphic function h_a on γ satisfying

(1) $F_a|_\gamma = F|_\gamma\, h_a$,
(2) $h_a(o) = 0$.

Then we have

$$\int_{\Delta'\times\Delta''} |F_a|^2 d\lambda_n \geq \frac{C_2}{|a|^2}$$

where C_2 is a positive constant independent of a, H_a and F_a.

3.2. Some topics may be benifit for the future research

3.2.1. Multiplier ideal sheaf $\mathcal{J}(\varphi)$

Definition 3 (Multiplier ideal sheaf)**.** Multiplier ideal sheaf $\mathcal{J}(\varphi)$ is considered as the subsheaf of germs of holomorphic functions f such that $|f|^2 e^{-2\varphi}$ is locally integrable (/summable).

It is worth to point out that it is a *coherent algebraic sheaf* over X and satisfies

$$H^q(X, K_X \otimes L \otimes \mathcal{J}(\varphi)) = 0$$

for all $q \geq 1$ if the curvature of L is positive in the sense of current.

The two efficient ways to solve the singularities are *Multiplier ideal sheaf* by Nadel (Which can be seen as a generalization of Kawamata-Viehweg's vanishing theorem.) and *The theory of positive currents* by Lelong. Currents can be seen as generalization of **algebraic cycles**.

3.2.2. Plurisubharmonic functions

Definition 4 (Plurisubharmonic functions)**.** A function $u : \Omega \longrightarrow [-\infty, +\infty[$ defined on an open subset $\Omega \subset \mathbb{C}^n$ is said to be plurisubharmonic (psh for short) if

(a) u is upper semicontinuous;

[30] A vital characteristic, which is equivalent to the integrability of the inegral.

(b) for every complex line $L \subset \mathbb{C}^n$, $u_{\restriction \Omega \cap L}$ is subharmonic on $\Omega \cap L$, that is, for all $a \in \Omega$ and $\xi \in \mathbb{C}^n$ with $|\xi| < d(a, \complement\Omega)$, the function u satisfies the mean value inequality[30]

$$u(a) \leqslant \frac{1}{2\pi} \int_0^{2\pi} u\left(a + e^{i\theta}\xi\right) d\theta.$$

The set of psh functions on Ω is denoted by $\mathrm{Psh}(\Omega)$.

Theorem 5. *A function $u \in C^2(\Omega, \mathbb{R})$ is psh if and only if the hermitian form*

$$Hu(a)(\xi) = \sum_{1 \leqslant j,k \leqslant n} \frac{{}^2u}{z_j \bar{z}_k(a)} \xi_j \bar{\xi}_k$$

is semi-positive at every point $a \in \Omega$.

3.2.3. Positive Currents

A current of degree q on an oriented differentiable manifold M is simply a differential q-form Θ with distribution coefficients. The space of currents of degree q over M will be denoted by $\mathscr{D}'^q(M)$. Alternatively, a current of degree q can be seen as an element Θ in the dual space $\mathscr{D}'_p(M) := (\mathscr{D}^p(M))'$ of the space $\mathscr{D}^p(M)$ of smooth differential forms of degree $p = \dim M - q$ with compact support; the duality pairing is given by

$$\langle \Theta, \alpha \rangle = \int_M \Theta \wedge \alpha, \quad \alpha \in \mathscr{D}^p(M). \tag{3.10}$$

A basic example is the current of integration $[S]$ over a compact oriented submanifold S of M :

$$\langle [S], \alpha \rangle = \int_S \alpha, \quad \deg \alpha = p = \dim_{\mathbb{R}} S \tag{3.11}$$

Then $[S]$ is a current with measure coefficients, and Stokes' formula shows that $d[S] = (-1)^{q-1}[S]$, in particular $d[S] = 0$ if S has no boundary. Because of this example, the integer p is said to be the dimension of Θ when $\Theta \in \mathscr{D}'_p(M)$. The current Θ is said to be closed if $d\Theta = 0$.

On a complex manifold X, we have similar notions of bidegree and bidimension; as in the real case, we denote by

$$\mathscr{D}'^{p,q}(X) = \mathscr{D}'_{n-p,n-q}(X), \quad n = \dim X, \tag{3.12}$$

the space of currents of bidegree (p, q) and bidimension $(n-p, n-q)$ on X. According to [2], a current Θ of bidimension (p, p) is said to be (weakly) positive if for every choice of smooth (1,0)-forms $\alpha_1, \ldots, \alpha_p$ on X the distribution

$$\Theta \wedge (i\alpha_1 \wedge \bar{\alpha}_1) \wedge \ldots \wedge (i\alpha_p \wedge \bar{\alpha}_p)$$

is a positive measure.

[2] P. Lelong. "Intégration sur un ensemble analytique complexe". In: *Bulletin de la Société Mathématique de France* 85 (1957), pp. 239–262. DOI: 10.24033/bsmf.1488. URL: http://www.numdam.org/articles/10.24033/bsmf.1488/

3.3. Perverse Sheaf and Intersection Cohomology

- test
  - testing
    - tested
  - testii
- test2
- test3

1. test
   (a). testing
      I. tested
   (b). testii
2. test2
3. test3



IV
Proof of Opennes Conjecture

4

Notes for the Proof of Openness Conjecture

4.1. Organizing the proof strategy

4.1.1. The case 1

Lemma 1 (Lemma 2.3 (see [8]). *; see also [22]).] Let $F \in \mathcal{O}_n$ and $g_1, \ldots, g_s \in \mathcal{O}_n$ be germs of holomorphic functions vanishing at the origin $o \in \mathbb{C}^n$. Assume that for any given neighborhood of o, $|F| \leq C\,|(g_1, \ldots, g_s)|$ does not hold for any constant C. Then there exists a germ of an analytic curve γ through o, satisfying $\gamma \cap \{F = 0\} \subseteq \{o\}$, such that $\frac{g_i}{F}\big|_\gamma$ is holomorphic on $\gamma \backslash o$ with 2 for any $i \in \{1, \ldots, s\}$, where $\frac{\widetilde{g_i}}{F}$ is the holomorphic extension of $\frac{g_i}{F}$ from $\gamma \backslash o$ to γ.*

Theorem 2 (cf [1]). *Let φ be a negative plurisubharmonic function on the unit polydisc $\Delta^n \in \mathbb{C}^n$. Suppose F is a holomorphic function on Δ^n, which satisfies*

$$\int_{\Delta^n} |F|^2 e^{-\varphi} d\lambda_n < +\infty,$$

where $d\lambda_n$ is the Lebesgue measure on $\mathbb{C}^n$. Then for some $r \in (0, 1)$, there exists a number $p > 1$ such that

$$\int_{\Delta_r^n} |F|^2 e^{-p\varphi} d\lambda_n < +\infty.$$

Proof. As

$$\int_{\Delta' \times \Delta''} |F_j|^2 e^{-p_j \varphi} d\lambda_n \leqslant C \int_{H_j} |F|^2 e^{-p_j \varphi} d\lambda_{n-1}$$

and we already known that[31]

$$\int_{H_j} |F|^2 e^{-p_j \varphi} d\lambda_{n-1} \leqslant 2 \int_{H_j} |F|^2 e^{-\varphi} d\lambda_{n-1} = o\left(\frac{1}{|a_j|^2}\right),$$

thus we have

$$\begin{aligned}\int_{\Delta' \times \Delta''} |F_j|^2 e^{-p_j \varphi} d\lambda_n &\leqslant C \int_{H_j} |F|^2 e^{-p_j \varphi} d\lambda_{n-1} \\ &\leqslant C \int_{H_j} |F|^2 e^{-\varphi} d\lambda_{n-1} = o\left(\frac{1}{|a_j|^2}\right).\end{aligned}$$

4.1 Organizing the proof strategy 37
4.1.1 The case 1 37
4.2 Some topics may be benifit for the future research 40
4.2.1 Multiplier ideal sheaf $\mathcal{J}(\varphi)$ 40
4.2.2 Plurisubharmonic functions 40
4.2.3 Positive Currents . . . 41
4.3 Perverse Sheaf and Intersection Cohomology . . . 42

[31] see (3.7) in original paper.

Then by $F_j|_{H_j} = F|_{H_j}$ and negativeness of φ, we have

$$\int_{\Delta'\times\Delta''} |F_j|^2 d\lambda_n \leqslant \int_{\Delta'\times\Delta''} |F_j|^2 e^{-p_j\varphi} d\lambda_n = o\left(\frac{1}{|a_j|^2}\right),$$

which is to say

$$\int_{\Delta'\times\Delta''} |F_j|^2 d\lambda_n = o\left(\frac{1}{|a_j|^2}\right).$$

We choose r_0 small enough such that[32]

$$\int_{\Delta_{r_0}} |F|^2 e^{-\varphi} d\lambda_1 < +\infty.$$

Step 1: Theorem 2 for the dimension one case. We first prove Theorem 2 for the dimension one case, which is elementary but instructive. Our proof for the general dimension case is quite similar. Actually the proofs for both cases are parallel.

We choose r_0 small enough such that $\int_{\Delta_{r_0}} |F|^2 e^{-\varphi} d\lambda_1 < +\infty$[33]. Then there exist complex numbers $a_j \to 0$ $(j \to +\infty)$ such that

$$\left|F\left(a_j\right)\right|^2 e^{-\varphi(a_j)} = o\left(\frac{1}{\left|a_j\right|^2}\right).$$

As $\left|F\left(a_j\right)\right|^2 e^{-\varphi(a_j)} < +\infty$, then one can find $p_j > 1$ small enough such that[34]

$$\left|F\left(a_j\right)\right|^2 e^{-p_j\varphi(a_j)} \leq 2\left|F\left(a_j\right)\right|^2 e^{-\varphi(a_j)} = o\left(\frac{1}{\left|a_j\right|^2}\right). \tag{4.1}$$

Using movable (respect to a_j) the Ohsawa-Takegoshi L^2 extension theorem on Δ, we obtain holomorphic functions F_j on Δ such that $F_j\big|_{a_j} = F\left(a_j\right)$ and[35]

$$\int_{\Delta} \left|F_j\right|^2 e^{-p_j\varphi} d\lambda_1 \leq \mathbf{C}\left|F\left(a_j\right)\right|^2 e^{-p_j\varphi(a_j)}, \tag{4.2}$$

where $\mathbf{C}$ is a universal constant.

By inequality (4.1) and negativeness of φ, we obtain that

$$\int_{\Delta} \left|F_j\right|^2 d\lambda_1 = o\left(\frac{1}{\left|a_j\right|^2}\right). \tag{4.3}$$

By contradiction, assume that Theorem 2 does not hold for $n = 1$; that is to say, $\int_{\Delta_r} |F|^2 e^{-p_j\varphi} d\lambda = +\infty$ for any $r > 0$ and any $j \in \{1, 2, \ldots\}$.

Assertion. Since $\{F = 0\} \cap \Delta_{r_0} \subseteq \{o\}$, then it follows from inequality (4.2) that one can derive that F/F_j *is unbounded*. Otherwise, the boundedness would imply the finiteness of the integral of $|F|^2 e^{-p_j\varphi}$, according to inequality (4.2). This contradicts the assumption. Then there exists a holomorphic function h_j on Δ_{r_0} satisfying

[32] The reason is that from the assumption of theorem 2, we have known that F is a holomorphic function on Δ^n, which satisfies

$$\int_{\Delta^n} |F|^2 e^{-\varphi} d\lambda_n < +\infty,$$

thus we can find a small enough r_0 such that

$$\int_{\Delta_{r_0}} |F|^2 e^{-\varphi} d\lambda_1 < +\infty.$$

[33] It has already been explained above.

[34] As

$$\begin{aligned} & e^{-p_j\varphi(a_j)} \\ &= [e^{-\varphi(a_j)}]^{p_j} \\ &= e^{-\varphi(a_j)} \cdot [e^{-\varphi(a_j)}]^{p_j-1}, \end{aligned}$$

$p_j > 1$ $(p_j - 1 > 0)$ and the negativeness of φ $(-\varphi > 0)$, there exists p_j small enough such that $p_j - 1 \leqslant \ln 2$, thus we have

$$[e^{-\varphi(a_j)}]^{p_j-1} \leqslant e^{p_j-1} \leqslant 2.$$

[35] This is because $|F(a_j)|^2 \cdot e^{-p_j(a_j)}$ is a constant.

(1) $F_j\big|_{\Delta_{r_0}} = F|_{\Delta_{r_0}} h_j$,
(2) $h_j(o) = 0$,
(3) $h_j\left(a_j\right) = 1$.

According to Lemma 2.1[36], it follows that

$$\frac{C_1}{\left|a_j\right|^2} \leq \int_{\Delta_{r_0}} \left|F_j\right|^2 d\lambda_1,$$

which contradicts equality (4.3), where $C_1 > 0$ is independent of j. We have thus proved Theorem 2 for $n = 1$.

Step 2: Theorem 2 for n. By induction on the dimension n, one may assume that Theorem 2 holds for $n - 1$.

We prove Theorem 2 for the general dimension n by contradiction. Assume that Theorem 2 for n is not true, therefore, for some negative psh function φ, there exists a holomorphic function F such that

$$\int_{\Delta_{r_0}^n} |F|^2 e^{-\varphi} d\lambda_n < +\infty, \tag{4.4}$$

for some $r_0 > 0$, and

$$\int_{\Delta_r^n} |F|^2 e^{-p\varphi} d\lambda_n = +\infty, \tag{4.5}$$

for any $r \in (0, r_0)$ and $p > 1$. That is to say, the germ of the holomorphic function F is in $\mathcal{I}(\varphi)_o$ but not in $\mathcal{I}_+(\varphi)_o$.

By Lemma 2.4[37] and equality 4.5, it follows that there exists a germ of an analytic curve γ through o satisfying $\left\{F|_\gamma = 0\right\} \subseteq \{o\}$ such that for any germ of holomorphic function g in $\mathcal{I}_+(\varphi)_o$, there exists a holomorphic function h_g on γ satisfying

$$h_g(o) = 0 \quad \text{and} \quad g|_\gamma = F|_\gamma h_g. \tag{4.6}$$

This plays a similar role with the assertion in the proof for dimension 1.

Using the local parametrization of γ (see [5]), without loss of generality, one may assume that γ and $\Delta' \times \Delta''$ are as in Section 2.1.

By inequality (4.4), it follows that there exist hyperplanes $H_j := H_{a_j} = \left\{z' = a_j\right\}$ that satisfy $a_j \to 0 (j \to \infty)$ and

$$\int_{H_j} |F|^2 e^{-\varphi} d\lambda_{n-1} = o\left(\frac{1}{\left|a_j\right|^2}\right). \tag{4.7}$$

According to *the induction assumption* and *the Lebesgue dominated convergence theorem*, it follows that for any given j, there exists a small enough $p_j > 1$ such that[38]

[36] Let $f \not\equiv 0$ be a holomorphic function on the disc Δ_r of radius r containing the origin o in $\mathbb{C}$. Let h_a be a holomorphic function on Δ_r, which satisfies $h_a(o) = 0$ and $h_a(b) = 1$ for any $b^k = a$ (k is a positive integer), where $a \in \Delta_r$ whose norm is small enough. Then we have

$$\int_{\Delta_r} |f|^2 |h_a|^2 d\lambda_1 > C_1 |a|^{-2}$$

where C_1 is a positive constant independent of a and h_a.

[37] Assume that $F \in \mathcal{O}_n$ is a holomorphic function on some neighborhood V of o that is not a germ of

$$\mathcal{I}\left(\psi_{j_1}\right)_o = \left(\cup_{j=1}^{\infty} \mathcal{I}\left(\psi_j\right)\right)_o.$$

Then there exists a germ of an analytic curve (γ, o) such that

$$\left.\frac{\widetilde{g \circ \gamma}}{F \circ \gamma}\right|_o = 0$$

holds for any germ (g, o) of $\mathcal{I}\left(\psi_{j_1}\right)_o$, where $\frac{\widetilde{g\circ\gamma}}{F\circ\gamma}$ is the holomorphic extension of $\frac{g\circ\gamma}{F\circ\gamma}$ from $\gamma \backslash o$ to γ.

[38] It may be the reason that from the truth of one case in (4.1), we can obtain that when $a_j \to 0$, we have (4.7), thus for any given j, by dominated convergence theorem, (4.8) is true.

$$\int_{H_j} |F|^2 e^{-p_j\varphi} d\lambda_{n-1} \leq 2\int_{H_j} |F|^2 e^{-\varphi} d\lambda_{n-1} = o\left(\frac{1}{|a_j|^2}\right). \tag{4.8}$$

Using movably (respect to $j \longrightarrow \infty$) *the Ohsawa-Takegoshi L^2 extension theorem* on $\Delta' \times \Delta''$, we obtain a holomorphic function $F_j := F_{a_j}$ on $\Delta' \times \Delta''$ for each j such that $F_j\big|_{H_j} = F|_{H_j}$, and

$$\int_{\Delta'\times\Delta''} \left|F_j\right|^2 e^{-p_j\varphi} d\lambda_n \leq \mathbf{C}\int_{H_j} |F|^2 e^{-p_j\varphi} d\lambda_{n-1},$$

where $\mathbf{C}$ is a universal constant. By inequality (4.8) and negativeness of φ, it follows that

$$\int_{\Delta'\times\Delta''} \left|F_j\right|^2 d\lambda_n = o\left(\frac{1}{|a_j|^2}\right) \tag{4.9}$$

Note that $F_j\big|_{H_j} = F|_{H_j}$ and $(F_j, o) \in \mathcal{I}_+(\varphi)_o$ but $(F, o) \notin \mathcal{I}_+(\varphi)_o$. According to equality (4.6) and Lemma 2.2[39], it follows that

$$\int_{\Delta'\times\Delta''} \left|F_j\right|^2 d\lambda_n \geq \frac{C_2}{|a_j|^2}$$

where $C_2 > 0$ is independent of j, which contradicts equality (4.9). We have thus proved Theorem 2 for n. The proof of Theorem 2 is now complete. □

[39] Let F be a holomorphic function on $\Delta' \times \Delta''$ satisfying $F|_\gamma \not\equiv 0$. Denote by hyperplane $H_a := \{z' = a\}$ near o. Let F_a be the holomorphic extension of $F|_{H_a}$ on $\Delta' \times \Delta''$ such that there exists a holomorphic function h_a on γ satisfying

(1) $F_a|_\gamma = F|_\gamma\, h_a$,
(2) $h_a(o) = 0$.

Then we have

$$\int_{\Delta'\times\Delta''} |F_a|^2 d\lambda_n \geq \frac{C_2}{|a|^2}$$

where C_2 is a positive constant independent of a, H_a and F_a.

4.2. Some topics may be benifit for the future research

4.2.1. Multiplier ideal sheaf $\mathcal{J}(\varphi)$

Definition 3 (Multiplier ideal sheaf)**.** Multiplier ideal sheaf $\mathcal{J}(\varphi)$ is considered as the subsheaf of germs of holomorphic functions f such that $|f|^2 e^{-2\varphi}$ is locally integrable (/summable).

It is worth to point out that it is a *coherent algebraic sheaf* over X and satisfies

$$H^q(X, K_X \otimes L \otimes \mathcal{J}(\varphi)) = 0$$

for all $q \geq 1$ if the curvature of L is positive in the sense of current.

The two efficient ways to solve the singularities are *Multiplier ideal sheaf* by Nadel (Which can be seen as a generalization of Kawamata-Viehweg's vanishing theorem.) and *The theory of positive currents* by Lelong. Currents can be seen as generalization of **algebraic cycles**.

4.2.2. Plurisubharmonic functions

Definition 4 (Plurisubharmonic functions)**.** A function $u : \Omega \longrightarrow [-\infty, +\infty[$ defined on an open subset $\Omega \subset \mathbb{C}^n$ is said to be plurisubharmonic (psh for short) if

(a) u is upper semicontinuous;

[40] A vital characteristic, which is equivalent to the integrability of the inegral.

(b) for every complex line $L \subset \mathbb{C}^n$, $u_{\restriction \Omega \cap L}$ is subharmonic on $\Omega \cap L$, that is, for all $a \in \Omega$ and $\xi \in \mathbb{C}^n$ with $|\xi| < d(a, \complement\Omega)$, the function u satisfies the mean value inequality[40]

$$u(a) \leqslant \frac{1}{2\pi} \int_0^{2\pi} u\left(a + e^{i\theta}\xi\right) d\theta.$$

The set of psh functions on Ω is denoted by $\mathrm{Psh}(\Omega)$.

Theorem 5. *A function $u \in C^2(\Omega, \mathbb{R})$ is psh if and only if the hermitian form*

$$Hu(a)(\xi) = \sum_{1 \leqslant j,k \leqslant n} \frac{\partial^2 u}{\partial z_j \partial \bar{z}_k}(a) \xi_j \bar{\xi}_k$$

is semi-positive at every point $a \in \Omega$.

4.2.3. Positive Currents

A current of degree q on an oriented differentiable manifold M is simply a differential q-form Θ with distribution coefficients. The space of currents of degree q over M will be denoted by $\mathscr{D}'^q(M)$. Alternatively, a current of degree q can be seen as an element Θ in the dual space $\mathscr{D}'_p(M) := (\mathscr{D}^p(M))'$ of the space $\mathscr{D}^p(M)$ of smooth differential forms of degree $p = \dim M - q$ with compact support; the duality pairing is given by

$$\langle \Theta, \alpha \rangle = \int_M \Theta \wedge \alpha, \quad \alpha \in \mathscr{D}^p(M). \tag{4.10}$$

A basic example is the current of integration $[S]$ over a compact oriented submanifold S of M :

$$\langle [S], \alpha \rangle = \int_S \alpha, \quad \deg \alpha = p = \dim_{\mathbb{R}} S \tag{4.11}$$

Then $[S]$ is a current with measure coefficients, and Stokes' formula shows that $d[S] = (-1)^{q-1}[S]$, in particular $d[S] = 0$ if S has no boundary. Because of this example, the integer p is said to be the dimension of Θ when $\Theta \in \mathscr{D}'_p(M)$. The current Θ is said to be closed if $d\Theta = 0$.

On a complex manifold X, we have similar notions of bidegree and bidimension; as in the real case, we denote by

$$\mathscr{D}'^{p,q}(X) = \mathscr{D}'_{n-p,n-q}(X), \quad n = \dim X, \tag{4.12}$$

the space of currents of bidegree (p, q) and bidimension $(n-p, n-q)$ on X. According to [2], a current Θ of bidimension $(\ p, p\ )$ is said to be (weakly) positive if for every choice of smooth $(\ 1{,}0\ )$-forms $\alpha_1, \ldots, \alpha_p$ on X the distribution

$$\Theta \wedge (i\alpha_1 \wedge \bar{\alpha}_1) \wedge \ldots \wedge (i\alpha_p \wedge \bar{\alpha}_p)$$

is a positive measure.

[2] P. Lelong. "Intégration sur un ensemble analytique complexe". In: *Bulletin de la Société Mathématique de France* 85 (1957), pp. 239–262. DOI: 10.24033/bsmf.1488. URL: http://www.numdam.org/articles/10.24033/bsmf.1488/

4.3. Perverse Sheaf and Intersection Cohomology

- test
  - testing
    - tested
  - testii
- test2
- test3

1. test
   (a). testing
      I. tested
   (b). testii
2. test2
3. test3

Bibliography

[1] Q. Guan and X. Zhou. “A proof of Demailly’s strong openness conjecture”. In: *Annals of mathematics* (2015), pp. 605–616.

[2] P. Lelong. “Intégration sur un ensemble analytique complexe”. In: *Bulletin de la Société Mathématique de France* 85 (1957), pp. 239–262. DOI: 10.24033/bsmf.1488. URL: http://www.numdam.org/articles/10.24033/bsmf.1488/.

5 Appendix

5.1 45
5.2 46
5.3 47

5.1.

- (https://mp.weixin.qq.com/s/ebhG6fFNryOkqKulTzFxeg)
-

5.2.

5.3.

函数/序列空间	C	C_0	C_{00}	$l^p\ (1\le p<\infty)$	l^∞	$C[a,b]$	$L^p\ (1\le p<\infty)$	L^∞
元素	$\{x_n\}\mid \lim_{n\to\infty} x_n<\infty\}$	$\{x_n\}\mid \lim_{n\to\infty} x_n=0\}$	$\{x_n\}\mid \exists N,\forall n>N, x_n=0\}$	$\{x_n\}\mid \sum_n \lvert x_n\rvert^p<\infty\}$	$\{x_n\}\mid \sup_n \lvert x_n\rvert<\infty\}$	$[a,b]$上的连续函数	$\{f\mid \int \lvert f\rvert^p d\mu<\infty\}$	$\{f\mid \text{ess-sup}\lvert f\rvert<\infty\}$
范数	$\sup_n \lvert x_n\rvert$			$(\sum_n \lvert x_n\rvert^p)^{\frac{1}{p}}$	$\sup_n \lvert x_n\rvert$	$\sup_{[a,b]}\lvert f\rvert$ / $\int_a^b \lvert f(x)\rvert dx$	$(\int \lvert f\rvert^p d\mu)^{\frac{1}{p}}$	$\text{ess-sup}\lvert f\rvert$
完备性	✓	✓	✗	✓	✓	✓ / ✗	✓	✓
可分性	✓	✓	✓	✓	✗	✓ (Weierstrass逼近定理)	✓	✗
可数稠子集	$\{(r_1,\dots,r_n,q,q,\dots)\mid r_n,q\in\mathbb{Q}\}_{n\in\mathbb{N}}$	$\{(r_1,\dots,r_n,0,\dots,0,\dots):r_1,\dots,r_n\in\mathbb{Q},n\in\mathbb{N}\}$			＼	有理系数多项式		＼
对偶空间	l^1	l^1	?	$\begin{cases} p=1: l^\infty \\ p>1: l^q,\ \frac{1}{p}+\frac{1}{q}=1 \end{cases}$	不是 l^1 ?	?	$\begin{cases} p=1: L^\infty \\ p>1: L^q,\ \frac{1}{p}+\frac{1}{q}=1 \end{cases}$	不是 L^1 ?
自反性 ($X\neq X^{**}$)	✗	✗	?	$\begin{cases} l^1: & ✗ \\ l^p\ (p>1): & ✓ \end{cases}$	✗	?	$\begin{cases} L^1: & ✗ \\ L^p\ (p>1): & ✓ \end{cases}$	✗
一些特殊的关系	C_0、C 是 l^∞ 的闭子空间 $\Rightarrow$ 是Banach空间. 若 $1\le p_2\le p_1<\infty$，且 $m(E)<\infty$，则 $L^\infty(E)\subset L^{p_1}(E)\subset L^{p_2}(E)$.							

1. **收敛（Convergence）**：一般指序列或函数列在某种意义下接近某个极限。具体形式取决于所考虑的拓扑或度量。
2. **一致收敛（Uniform Convergence）**：函数列 $\{f_n\}$ 在集合 S 上一致收敛到函数 f，如果对于任意的 $\epsilon > 0$，存在正整数 N，使得当 $n \geq N$ 时，对于所有 $x \in S$，都有 $|f_n(x) - f(x)| < \epsilon$。一致收敛意味着收敛速度在 S 上是一致的。
3. **点点收敛（Pointwise Convergence）**：函数列 $\{f_n\}$ 在集合 S 上点点收敛到函数 f，如果对于每个 $x \in S$，序列 $\{f_n(x)\}$ 收敛到 $f(x)$。点点收敛只关心每个点的极限，不考虑收敛速度。
4. **几乎处处收敛（Almost Everywhere Convergence）**：函数列 $\{f_n\}$ 在集合 S 上几乎处处收敛到函数 f，如果存在 S 的零测度子集 E，使得 $\{f_n\}$ 在 $S \setminus E$ 上点点收敛到 f。几乎处处收敛是点点收敛的弱化，允许在零测度集上不收敛。
5. **依测度收敛（Convergence in Measure）**：函数列 $\{f_n\}$ 在集合 S 上依测度收敛到函数 f，如果对于任意的 $\epsilon > 0$，$\{x \in S : |f_n(x) - f(x)| \geq \epsilon\}$ 的测度趋于 0。依测度收敛比几乎处处收敛更弱，只关心不收敛的点集的测度。
6. **平方平均收敛（Convergence in Mean）**：函数列 $\{f_n\}$ 在集合 S 上平方平均收敛到函数 f，如果 $\int_S |f_n(x) - f(x)|^2 \, dx \to 0$。平方平均收敛是 L^2 空间中的收敛形式。
7. **强收敛（Strong Convergence）**：在巴拿赫空间中，序列 $\{x_n\}$ 强收敛到 x，如果 $\|x_n - x\| \to 0$。强收敛是赋范空间中的收敛形式，与范数有关。
8. **弱收敛（Weak Convergence）**：在巴拿赫空间中，序列 $\{x_n\}$ 弱收敛到 x，如果对于每个连续线性泛函 f，都有 $f(x_n) \to f(x)$。弱收敛比强收敛更弱，只关心序列在连续线性泛函下的像的收敛。